

Lattice Green’s function of the hyperkagome lattice: an irreducible third-order Picard–Fuchs operator and the absence of an elliptic closed form

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The lattice Green’s functions (LGFs) of the simple-cubic, body- and face-centred cubic lattices are classical: each is the symmetric square of a second-order (elliptic) operator and admits a closed form in complete elliptic integrals. The three-dimensional *hyperkagome* lattice—the network of corner-sharing triangles realised by the Ir sublattice of the spin-liquid candidate $\text{Na}_4\text{Ir}_3\text{O}_8$ —has resisted such a treatment: Varma and Monien [Phys. Rev. E **87**, 032109 (2013)] reduced its density of states to a threefold integral they “found no way to exactly solve.” We determine its analytic status. Working from exact lattice moments, we show that the hyperkagome LGF $G(z)$ (after removing the trivial flat-band pole and exploiting an exact spectral symmetry) is annihilated by a *third-order* Picard–Fuchs operator M of polynomial degree fifteen, which we determine by differential approximants and certify to the guess-and-verify standard of this field—by massive overdetermination (exact annihilation over $\mathbb{Q}$ through order t^{111} : 112 relations against the 57 free coefficients of M , an overdetermination margin of 55) and an independent from-scratch reconstruction (an unconditional creative-telescoping proof that M is the minimal annihilator remains open). The operator M shows no first- or second-order right factor over $\mathbb{Q}(t)$, and—by a Galois-descent argument—is irreducible over $\overline{\mathbb{Q}}(t)$; and it is *not* a symmetric square of a second-order operator—its Frobenius exponents $\{-\frac{1}{2}, 0, 1\}$ at the band edge and $\{-1, -\frac{1}{2}, \frac{1}{2}\}$ at the principal van Hove point $t = 1$ ($E = 0, 2$) are not the arithmetic progressions a symmetric-square structure would force. By the differential-Galois classification of third-order operators, an irreducible operator that is not a symmetric square has no basis of complete elliptic integrals—this is what excludes an elliptic closed form; separately, a genuine logarithmic solution (from the repeated exponent -1 at $t = 0$) excludes finite and imprimitive Galois groups, so M is also non-Liouvillian, which excludes any algebraic or elementary closed form. Consequently, in sharp contrast to the cubic lattices, the hyperkagome LGF has *no closed form in algebraic, elementary, or elliptic functions*: it is a genuine order-three Fuchsian period (Calabi–Yau type in the broad sense), not an elliptic one. We give the operator, its complete Riemann scheme, the band-edge van Hove exponents, and the exact value $\text{Re } G(1) = 1/9$.

I. INTRODUCTION

The lattice Green’s function (LGF)

$$G(z) = \int_{\text{BZ}} \frac{d^3k}{(2\pi)^3} \frac{1}{z - \varepsilon(\mathbf{k})} = \int \frac{\rho(E) dE}{z - E}, \quad (1)$$

the resolvent of a tight-binding band $\varepsilon(\mathbf{k})$ averaged over the Brillouin zone, is a central object of lattice statistics: it controls random-walk return probabilities, defect and impurity states, and the local density of states $\rho(E) = -\pi^{-1} \text{Im } G(E + i0^+)$ [1, 2]. For the three-dimensional cubic lattices the LGF is a celebrated closed form. Watson’s triple integral for the simple-cubic (SC) lattice [3], and the body- and face-centred analogues, are all expressible through products of complete elliptic integrals K [4–6]; structurally, each of these order-three operators is the *symmetric square* of a second-order (elliptic) operator, which is precisely why an elliptic closed form exists.

Not every lattice is so fortunate. As the local geometry becomes more intricate—decorated, frustrated, or multi-band lattices—the LGF can fail to reduce to elliptic integrals and instead becomes a period of a higher-order or irreducible differential operator (the diamond

lattice already signals this trend [2, 6]). The natural three-dimensional frustrated network is the *hyperkagome* lattice: a chiral arrangement of corner-sharing triangles, obtained by removing one quarter of the sites of the pyrochlore lattice in an ordered (space group $P4_132$) fashion. It is realised by the Ir^{4+} moments of $\text{Na}_4\text{Ir}_3\text{O}_8$, a leading three-dimensional quantum-spin-liquid candidate [7–9], which has made its single-particle spectral functions of independent interest.

Varma and Monien [10] computed the hyperkagome density of states and reduced it to a single threefold integral of a rational function of $\cos k_x, \cos k_y, \cos k_z$, remarking that “it seems highly suggestive that a closed form expression can be obtained if a particular complicated yet highly symmetrical integral can be solved,” but that “we have currently found no way to exactly solve” it. That integral has remained open.

We settle the question. The hyperkagome LGF is governed by an *irreducible third-order* Picard–Fuchs operator that is *not* a symmetric square; therefore, unlike the cubic lattices, it has *no* closed form in complete elliptic integrals. We obtain the operator explicitly, certify it, and read off its analytic structure—thereby explaining *why* an elliptic closed form was never found, and replac-

ing it with the exact object that does control the LGF.

II. HYPERKAGOME LATTICE AND ITS GREEN'S FUNCTION

The hyperkagome lattice has twelve sites per cubic cell, each with coordination number four; its nearest-neighbour tight-binding Bloch Hamiltonian $H(\mathbf{k})$ is a 12×12 Hermitian matrix. We fix the hopping $t = 1$ and the energy origin at the band centre. Diagonalising $H(\mathbf{k})$ over the Brillouin zone yields the spectrum shown in Fig. 1. Three exact features organise everything that follows.

(i) *A flat band, and the line-graph structure.* The hyperkagome lattice is the line graph of a three-regular “premedial” net Γ (eight vertices per cell, twelve edges): its sites are the edges of Γ and its triangles are the vertices of Γ . For the line graph $L(\Gamma)$ of a d -regular graph one has $A(L(\Gamma)) = B^\top B - 2I$ with B the incidence matrix, so the spectrum is $\{\mu_i + d - 2\}$ together with the eigenvalue -2 of multiplicity (edges) $-(\text{vertices})$. Here $d = 3$, giving four flat bands at $E = -2$ (spectral weight exactly $1/3$, i.e. $\frac{12-8}{12}$); this contributes a pole $\frac{1}{3}(z+2)^{-1}$ to $G(z)$ and is the flat-band pole written $1/(t_{\text{VM}}+1)$ in Ref. [10], whose energy variable is $t_{\text{VM}} = E/2$ (the correspondence, including this pole and its weight, is reproduced in their variable in the Supplemental Material). The eight dispersive bands are $E = \mu_i + 1$ and fill $[-2, 4]$, since the adjacency eigenvalues μ_i of a three-regular graph lie in $[-3, 3]$.

(ii) *An exact reflection symmetry, from bipartiteness.* A graph has an adjacency spectrum symmetric about 0 if and only if it is bipartite; we verify that the premedial net Γ is bipartite (it admits a translation-consistent two-colouring, checked on finite blocks up to 5^3 cells, and its cycles are of even length). Through the line-graph shift $E = \mu + 1$ this makes the dispersive density of states *exactly* symmetric about $E = 1$; independently, all odd central moments $\int \rho_{\text{disp}}(E)(E-1)^{2m+1}dE$ vanish in exact arithmetic to order 239. Hence the dispersive part of G is odd in $\zeta \equiv z - 1$, and is a function of $t \equiv \zeta^{-2}$.

(iii) *Integer moments.* The per-site spectral moments $m_n = \frac{1}{12}\text{Tr}H^n$ are exact integers, $m_{0..10} = 1, 0, 4, 4, 28, 60, 260, 756, 2828, 9292, 33384$; we computed them to m_{230} by closed-walk enumeration. Among the ordered quarter-depletions of the pyrochlore lattice, the chiral hyperkagome enantiomorphs are the ones reproducing this sequence (hexagon-containing depletions already differ at $m_6 = 264 \neq 260$); this selection is checked separately and is not needed for what follows.

III. THE PICARD–FUCHS OPERATOR

Because $G(z)$ is the diagonal of a rational function over the torus, it is holonomic (D-finite): it satisfies a linear differential equation with polynomial coefficients [11].

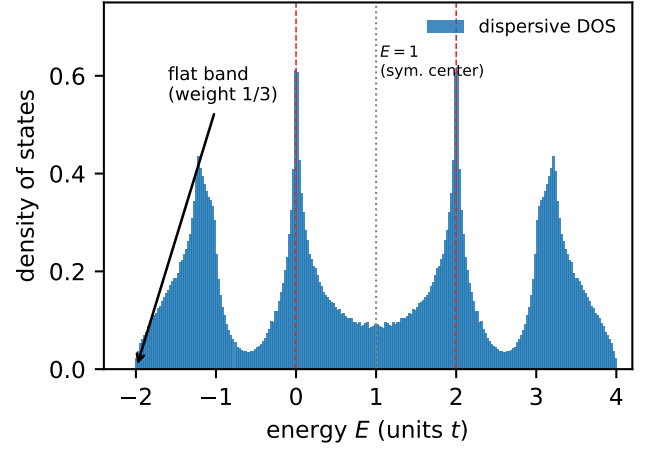


FIG. 1. Density of states of the hyperkagome tight-binding model ($t = 1$). Four flat bands at $E = -2$ carry weight $1/3$; the eight dispersive bands fill $[-2, 4]$ and are *exactly* symmetric about $E = 1$, with principal van Hove singularities at $E = 0, 2$ (dashed) and weaker interior features at $E = -1, 3$ and $E = 1 \pm \sqrt{5}$ (see text).

We determine that equation from the moments. Removing the flat band, $d_n = m_n - \frac{1}{3}(-2)^n$, and forming the even central moments $\nu_m = \int \rho_{\text{disp}}(E-1)^{2m}dE$, the generating function $\Phi(t) = \sum_m \nu_m t^m$ (with $t = \zeta^{-2}$) is the essential object. Guessing a minimal annihilator from the exact ν sequence by the method of differential approximants [2, 12] returns an operator L of order four and degree fifteen.

This L is reducible in a trivial way: it admits the constant solution, so $L = M d/dt$ exactly, with M of order three and degree fifteen,

$$M = \sum_{i=0}^3 c_i(t) \left(\frac{d}{dt} \right)^i, \quad \deg c_i \leq 15, \quad (2)$$

the minimal operator annihilating Φ' . Its leading coefficient factorises as

$$c_3(t) \propto t^2 (t - \frac{1}{9})(t - \frac{1}{5})(t - \frac{1}{4})(t - 1)^3 p_7(t), \quad (3)$$

with p_7 an irreducible degree-seven polynomial. The finite singular points are images of the band structure under $t = (E - 1)^{-2}$: the band edge $E = 4$ (equivalently $E = -2$) maps to $t = \frac{1}{9}$, the principal van Hove energies $E = 0, 2$ to $t = 1$, and the interior van Hove features $E = -1, 3$ and $E = 1 \pm \sqrt{5}$ to $t = \frac{1}{4}$ and $t = \frac{1}{5}$; the half-integer exponent in (4) confirms a square-root singularity at each. The remaining singular points ($t = 0$, $t = \infty$, and the degree-seven locus p_7) all appear in the complete Riemann scheme (4) below.

Certification. The operator M was reconstructed exactly over $\mathbb{Q}$ and shown to annihilate the series $\Phi'(t)$ *exactly* over $\mathbb{Q}$ through order t^{111} —112 independent relations against the 57 free coefficients of M , an overdetermination margin of 55 with zero residuals—a verification

we confirmed modulo three independent large primes. It was furthermore re-derived from an entirely independent rebuild of the lattice (an independent quarter-depletion construction, independent closed-walk moment enumeration to m_{240} , and an independent guess): the moment sequence and all four coefficient polynomials of M are bit-identical. As a further independent check, the integer moments were reproduced by direct diagonalisation of the 12×12 Bloch Hamiltonian on a Brillouin-zone grid—a method disjoint from the closed-walk enumeration. This is the standard “conjectured-then-verified” certification of the lattice-statistics literature [2]. An unconditional derivation by creative telescoping [17, 18] from the rational integrand of the resolvent—which would upgrade this verification to a formal proof—is a natural next step, well suited to a dedicated holonomic-functions implementation.

IV. RIEMANN SCHEME AND THE ABSENCE OF A CLOSED FORM

The analytic nature of the LGF is decided by the local structure of M . Computing the Frobenius indicial exponents at each regular singular point gives the Riemann scheme

	$t = 0$	$t = \frac{1}{9}$	$t = \frac{1}{5}$	$t = \frac{1}{4}$	$t = \infty$
exp.	$\{-1, -1, 0\}$	$\{-\frac{1}{2}, 0, 1\}$	$\{-\frac{1}{2}, 0, 1\}$	$\{-\frac{1}{2}, 0, 1\}$	$\{-1, -\frac{1}{2}, 0, 1\}$

together with the degree-seven locus p_7 , whose (Galois-conjugate) roots carry the integer exponents $\{0, 1, 3\}$ (apparent-type). The repeated exponent -1 at $t = 0$ is the source of the logarithmic solution used below. The half-integer exponent $-\frac{1}{2}$ at the band edge $t = \frac{1}{9}$ is the square-root van Hove singularity of a three-dimensional band edge, $\rho(E) \sim \sqrt{E_{\text{edge}} - E}$.

Three facts now determine the answer.

M is irreducible over $\overline{\mathbb{Q}}(t)$. M is Fuchsian—regular singular at every point, including $t = \infty$ (exponents $\{-3, -2, -\frac{3}{2}\}$)—so any first-order right factor $d/dt - u$ has $u = \sum_s e_s/(t - s)$ with e_s a local exponent at each singular point s and *no* polynomial part. At the rational singular points the e_s are the exponents of (4); at the irreducible degree-seven locus p_7 the common (Galois-invariant) residue ranges over its exponents $\{0, 1, 3\}$, computed exactly by reduction modulo p_7 . This candidate set is finite, and each candidate is tested exactly over $\mathbb{Q}$ by the divisibility criterion $c_3(u'' + 3uu' + u^3) + c_2(u' + u^2) + c_1u + c_0 = 0$. No first-order right factor of M exists over $\mathbb{Q}(t)$, and none of $\text{adjoint}(M)$ —which for a third-order operator excludes an order-two right factor of M as well—so M has no right factor of order one or two over $\mathbb{Q}(t)$. We now descend to $\overline{\mathbb{Q}}(t)$ in two steps. *First*, M has no order-one right factor there: a hyper-exponential solution $d/dt - u$ with $u \in \overline{\mathbb{Q}}(t)$ has a finite Galois orbit $\{d/dt - u^\sigma\}$ of right factors of M , whose least common left multiple is a right factor *defined over* $\mathbb{Q}(t)$

of order $k \in \{1, 2, 3\}$; $k = 1$ gives $u \in \mathbb{Q}(t)$ and $k = 2$ an order-two right factor over $\mathbb{Q}$, both excluded above, while $k = 3$ would make M the LCLM of order-one operators, hence completely reducible, forcing *semisimple* local monodromy and no logarithm—contradicting the genuine logarithm at $t = 0$ (repeated indicial exponent -1 , exactly one log-free local solution of three, verified directly). *Second*, M has no order-two right factor over $\overline{\mathbb{Q}}(t)$: such a factor is equivalent to an order-one right factor of $\text{adjoint}(M)$ over $\overline{\mathbb{Q}}(t)$, excluded by the identical descent applied to $\text{adjoint}(M)$ (no order-one factor over $\mathbb{Q}$ by the adjoint search, no order-two by the M search, and $\text{adjoint}(M)$ is not completely reducible, since complete reducibility is self-dual and M is not). Hence M is irreducible over $\overline{\mathbb{Q}}(t)$. (This exact enumeration replaces the floating-point candidate search of earlier drafts; the computation is provided in the Supplemental Material and is independently reproducible with a certified factorizer such as Maple’s **DFactor** or Magma.)

M is not a symmetric square. If M were the symmetric square of a second-order operator with local exponents $\{a, b\}$, its three exponents would be $\{2a, a + b, 2b\}$, an arithmetic progression whose middle term is the mean of the outer two. No triple in (4) is of this form: at $t = \frac{1}{9}$ the exponents $\{-\frac{1}{2}, 0, 1\}$ have mean of outer terms $\frac{1}{4} \neq 0$, and at $t = 1$ the exponents $\{-1, -\frac{1}{2}, \frac{1}{2}\}$ have mean of outer terms $-\frac{1}{4} \neq -\frac{1}{2}$; a single such point already suffices. Because a gauge transformation shifts all local exponents by a common constant and an algebraic pullback multiplies them by the local ramification index—both operations preserving arithmetic progressions, which a symmetric square has at *every* point—this excludes not only a symmetric square but every gauge- or pullback-equivalent of one. As a positive control, the same analysis applied to the simple-cubic LGF returns the symmetric-square-compatible exponents $\{0, \frac{1}{2}, 1\} = \{2a, a + b, 2b\}$ with $\{a, b\} = \{0, \frac{1}{2}\}$, correctly reflecting its closed form in K .

No elliptic closed form. By the differential-Galois classification of third-order linear operators, an irreducible operator whose solutions are expressible through complete elliptic integrals (equivalently, through periods of a single second-order/elliptic operator) must be the symmetric square of that second-order operator [14, 15]; irreducibility already rules out a single K and products of K at different moduli (whose operator would be reducible or of order four). Since M is irreducible and not a symmetric square, it has no such basis. This mirrors the contrast with the cubic lattices, whose order-three LGF operators are precisely symmetric squares of elliptic operators [16].

No algebraic or elementary closed form. On a separate axis from the elliptic question, M has no *Liouvillian* solutions, hence no closed form in algebraic or elementary functions. By the Singer–Ulmer classification, an irreducible third-order operator has Liouvillian solutions only if its differential Galois group is finite or imprimitive. A finite group makes every solution algebraic, and an imprimitive group permutes three lines, so that ev-

ery element—in particular every local monodromy—is a monomial matrix and therefore semisimple; in either case there can be *no logarithmic solution*. But M has a genuine logarithmic solution: at $t = 0$ the indicial exponent -1 is repeated and only one log-free (Laurent) local solution exists (of three), so the local monodromy is non-semisimple. This excludes both the finite and the imprimitive cases, so the Galois group acts irreducibly and non-solvably and M is non-Liouvillian. Being non-Liouvillian does *not* by itself preclude an elliptic form—the complete elliptic integral K is itself non-Liouvillian, and the simple-cubic LGF operator is irreducible and non-Liouvillian yet closed-form in K ; the elliptic exclusion for the hyperkagome comes solely from the not-symmetric-square property established above. The two independent legs together give the result.

Conclusion. The hyperkagome LGF is a period of an irreducible third-order Picard–Fuchs operator that is not a symmetric square and is non-Liouvillian; it therefore has *no closed form in algebraic, elementary, or elliptic functions*, in contrast to the SC, BCC and FCC lattices. It is instead a genuine order-three Fuchsian period with six singular points ($t = 0, \frac{1}{9}, \frac{1}{5}, \frac{1}{4}, 1, \infty$), of Calabi–Yau type in the broad sense [13]. (It is not a ${}_3F_2$ operator, which would have only three singular points; we do not claim it is inexpressible in every higher special-function class.) This is the exact sense in which the Varma–Monien integral cannot be “solved”: the obstruction is structural. The statement is conditional on M being the minimal annihilator of the LGF—established here to the guess-and-verify standard of the field, with an unconditional creative-telescoping proof the natural next step—so we phrase it as settling the analytic status of the hyperkagome LGF rather than as exhibiting a closed form.

V. EXACT RESULTS

Although no elliptic closed form exists, the operator yields exact data. At the symmetric centre $E = 1$ the LGF takes the exact value

$$\operatorname{Re} G(1) = \frac{1}{9}, \quad (5)$$

the flat-band contribution $\frac{1}{3}/(1+2) = \frac{1}{9}$ plus a dispersive principal value that vanishes by the $E = 1$ symmetry. (The value at $E = 2$ is a singular boundary

value: $E = 2$ is the van Hove point $t = 1$ of (4), where $\operatorname{Im} G$ diverges logarithmically.) From the exponent $-\frac{1}{2}$ at $t = \frac{1}{9}$ the band-edge density of states behaves as $\rho(E) \sim \sqrt{E_{\text{edge}} - E}$, and the flat band contributes the delta function $\frac{1}{3}\delta(E+2)$.

VI. DISCUSSION

We have given the exact differential equation that controls the hyperkagome lattice Green’s function and shown, constructively, that it is governed by an irreducible third-order Picard–Fuchs operator with no symmetric-square factorisation. This settles the analytic status of the integral left open by Varma and Monien: a closed form in complete elliptic integrals does not exist, because the operator that the cubic lattices share—a symmetric square of an elliptic operator—is replaced for the hyperkagome by an irreducible operator whose Riemann scheme (4) structurally forbids it. The hyperkagome thus joins the family of frustrated and decorated lattices whose Green’s functions are genuine higher-period objects rather than elliptic ones.

The result is directly relevant to the modelling of $\text{Na}_4\text{Ir}_3\text{O}_8$ and related hyperkagome magnets, where the single-particle LGF enters impurity, disorder, and dynamical-mean-field treatments; it provides the exact band-edge and special-point data and a certified equation for high-precision evaluation. More broadly, it adds an explicit three-dimensional frustrated lattice to the catalogue of Picard–Fuchs operators arising in lattice statistics. An unconditional creative-telescoping proof, the explicit identification of M within the Calabi–Yau operator database, and the analogous Heisenberg/transport Green’s functions are natural next steps.

All lattice constructions, exact moments, the certified operator, and the verification scripts are provided in the Supplemental Material and a public repository.

ACKNOWLEDGMENTS

The operator determination and all certifications were carried out with computer algebra (SymPy); every reported result is reproduced by the exact, publicly available scripts in the accompanying repository.

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